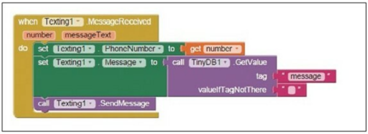

Figure 4: Block arrangement for SMS received

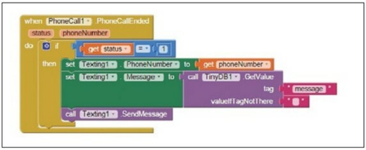

Figure 5: Block arrangement for sending SMS

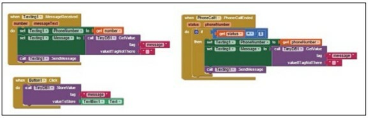

Figure 6: The complete picture

database, we have to assign it a tag so that it can be retrieved easily. So we have stored our text with the tag name *message*.

2. The block shown in Figure 4, shows that when any incoming SMS has been received, it will extract the attached number and will fetch the pre-saved message from the database.
3. When it has a number and message, it will call the *send message method* (Figure 5) which will deliver the message to the correct recipient.

The block gets active when the incoming call has ended and it has not been answered. Here the status of the check point is 1, which shows that incoming call was not picked. Hence it is a missed call.

Likewise, the previous block will get the phone number and pre-saved message from the database and will call the *send message method* to deliver the text to the correct recipient.

Figure 6 shows how the block editor looks like with all the blocks put together in the block editor.

Packaging and testing

To test your first android application, you need to get it on your android phone. You have to download the application to your computer and then move it to your phone via Bluetooth or USB cable. The steps to download your newly built application are mentioned below.

1. On the top row, click on the *Build* button. It will show you the option to download the *apk* to your computer.
2. While downloading, you can view its progress and after it has been successfully done, the application will be placed in the download folder of your directory or the location you have set for your downloads.
3. Now you need to get this *apk* file to your mobile phone either via Bluetooth or via USB cable. Once you have placed the *apk* file to your SD card, you need to install it. Follow the on-screen instructions to install it. You might get some notification or warning saying "Installation from an un-trusted source". Allow this from the settings of your phone. After successful installation you will see your application's icon in the menu of your mobile. Here you will see the default icon which can be changed and you will learn how as we move ahead in this series.

Your application should work exactly as per the specifications you have given. Now depending on your needs, you can change various things like images, sound and behaviour.

Debugging the application

We have just created the prototype of the application with very basic functionality and what most users might be interested in. Now let's look at various aspects that require serious attention so as not to annoy the user. Consider the following cases:

1. What will happen if the user has not stored any text?
2. What if the user has typed text in the text box but not pressed the *Save* button. Will the app be able to send text? Will it be able to send an SMS when the app is opened the next time?
3. If we want to respond only to particular contacts, then how can we filter the numbers from incoming calls or messages?

These are some creepy scenarios that could occur and a user would want your application to respond to the situation with intelligence

Think over these scenarios and figure out how you can integrate them into the application? Do ask me if you fail to address any of the above cases. END Faciaest, te volo

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